Starting as overdensities in the initial density fluctuations and evolving into the largest gravitationally bound structures in the Universe, galaxy clusters are a rich testbed of the cosmos. Their large masses, self similarity, combination of both luminous and dark matter, and multi-wavelength observables, make them ideal candidates for studying cosmology. Specifically, galaxy clusters allow us to probe the fundamental nature of dark energy through understanding the growth of structure in the Universe.

In order to set competitive limits, a number of systematics must be studied and ideally mitigated, like those induced by optical cluster selection, shear and redshift calibration from blending, and model misspecification. Over the next few years, I plan on both performing early science measurements on clusters from Roman, Rubin, and Euclid along with working on novel methods of mitigation that can be implemented in future data releases. My previous experiences with cluster, shear calibration, and on-going membership of LSST DESC and Rubin commissioning team equip me with the necessary tools to successfully carry out this research.

**Previous and Current Research** *Blending* During my PhD, I focused on the weak lensing systematic of **unrecognized blends** which are objects (typically galaxies) that are mistakenly identified as a single entity due to severe amounts of overlap between sources (see Fig. 1 for an example). The repercussions of unrecognized blends via detection bias alone have shown shifts in cosmology parameters at the level of a few percent [**ben˙clustering**, **blends˙2011**, **kidsblends˙2025**], and an early study including the impact on shape and flux measurements on cosmic shear found up to a $2\sigma$ shift in $S_8$ [**erfan**].

For 3x2pt cosmological analysis, expensive image simulation is used to measure the bias induced by unrecognized blends on redshift and shear calibration. Directly identifying unrecognized blends at the catalog level is an appealing solution that I pursued during my PhD.

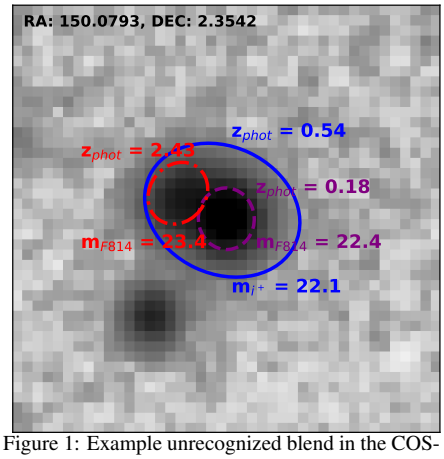


Figure 1: Example unrecognized blend in the COSMOS field with observed magnitudes and estimated photometric redshifts attached. Dark blue is the object as detected and measured by Subaru, dashed-dotted red and purple are detections and measurements from HST.

In [**unrec˙liang**], we showed that unrecognized blends can be readily detected at the catalog level using machine learning techniques that assign an unrecognized blend probability based on measured fluxes. While we showed that **it is possible to remove unrecognized blends from the catalog using observational data alone**, it was unclear if such a removal would ultimately improve our cosmology. Previous work suggested that it would, but there are a number of assumptions being made in those works that limit the applicability to LSST cosmology. Instead of directly measuring cosmological contours, I investigated the efficacy of our blend removal method on shear calibration using the AnaCal shear framework to decouple the effect of blending on redshift and shear. We showed that not only does the bias reduce, **the bias can be almost totally removed** with a 30-40% sample cut [**unrec˙shear**]. This is a promising alternative mitigation strategy that can totally remove a large systematic bias that LSST will suffer from.

*Clusters* I lead the LSST DESC Project 408 "A Rubin view of Abell 360", a collaborative effort between the Rubin Observatory and DESC, to study the galaxy cluster Abell 360 on commission-

ing data. This is some of the first weak lensing results to come from the Rubin Observatory and acts as proof that the instrument is able to deliver science quality data even in suboptimal conditions. I was involved with **all steps of analysis** from red sequence identification to mass fitting, specifically focused on **source sample selection, photo-$z$ validation, shear measurement and calibration with shapeHSM and AnaCal, cluster mass fitting**, and helped other collaborators create aperture mass maps, study 1D galaxy distributions, and evaluate cuts with synthetic source injection (SSI). This was my first foray into clusters along with leading a large collaborative effort, both of which were excellent learning experiences for me. Leading scientific discussion, coordinating paper writing efforts, and organizing almost daily telecons were all necessary skills to complete this work.

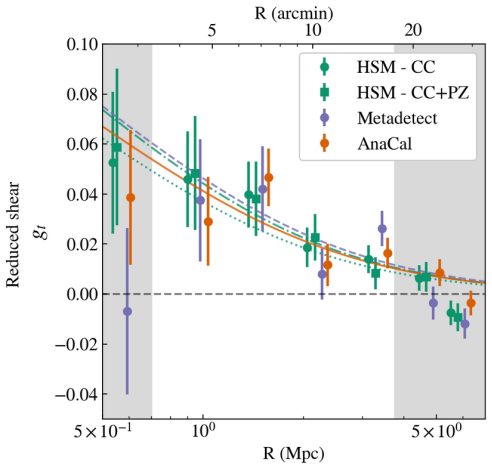



Figure 2: Tangential shear signal measured around Abell 360 using a variety of shape measurements and source selections. [paper in prep.]

Following my efforts on Abell 360, I have been working on testing the combination of the cutting edge shear (AnaCal) and photo-$z$ (RAIL) estimators by measuring cluster profiles in the Euclid Deep Field South [publication in prep.] In this work we have **measured tangential shear signals around clusters that were undetectable in the base Rubin catalog** due to cleaner sample selection and higher quality shear estimates, highlighting the power of the combination of AnaCal and RAIL. The combination of these tools is currently being used for HSC Year 6 cosmic shear and **this is the first cluster analysis** using AnaCal+RAIL.

Throughout my PhD, I have worked extensively on understanding the image level processing of data focusing almost entirely on the Rubin observatory. Along the way I have developed new weak lensing mitigation strategies [**unrec˙liang**, **unrec˙shear**], produced some of the first weak lensing results from Rubin [**a360**], and have a familiarity with the tools used in modern cosmological analysis [**edfs**].

**Research Proposal** Galaxy clusters are powerful probes into the growth of structure in the universe but can suffer from several sources of bias. Some of the largest sources of biases can be mitigated by using an X-ray or SZ selection but require a tighter understanding of weak lensing systematics. Sub-percent level constraints on $\sigma_8(z)$ are achievable for LSST if the mass scatter can be constrained to within $\pm0.05$[heidi] whereas the latest cluster calibrations are limited to $\sim 0.2$ [bocquet, grandis]. The majority of that error budget comes from redshift calibration which is impacted by blending. Furthermore, it is **currently unknown if modern shear calibrations, AnaCal and Metadetect, are able to meet the 1% shear calibration requirement needed for cluster weak lensing when accounting for realistic cluster conditions**. Accurately measuring the levels of bias for cluster weak lensing is critical to obtain precision cosmology and these measurements will directly motivate future developments for cluster specific measurement tools. Over the next X years, I intend to focus on validating the tools used for cluster weak lensing and then measure cluster lensing profiles with Roman and Rubin.

*Goal 1: Cluster measurements for Roman and Rubin*

Precise ensemble measurements of galaxy clusters require an estimate of both statistical and systematic bias. For clusters, this is traditionally done with N-body or hydrodynamic simulations that can capture physics-induced bias (like miscentering of optical clusters due to triaxiality), but are computationally expensive and can ignore biases coming from image processing. I propose to work on both direct image generation and synthetic source injection (SSI) to validate shear and redshift measurements coming from science pipelines.

Direct image generation has been the standard technique to calibrate any biases in cosmic shear [macrann, hsc] as it is able to directly test for systematic biases inherent in the tools and algorithms being used. Incorporating a redshift varying background population, foreground cluster members and contaminants, intra-cluster light, cluster triaxiality, and more realistic sky background variation are some of the improvements needed to properly test clusters. Adding these systematics into our simulations sequentially will allow us to understand the bias introduced by each feature to inform future work on mitigation strategites.

While image generation is a powerful tool, it relies on the correct understanding and implementation of all the systematics that exist in data. Alternatively, SSI allows us to characterize the existing data and test the measurement pipeline directly on synthetic sources. In previous works, SSI has been used to validate flux measurements and star-galaxy separation [add citations] and when studying Abell 360, this technique was used to understand the effect of various cuts on the source sample. Repeating this work for Roman and Rubin clusters on a larger scale will allow us to validate measurements made on the real data.

Performing both image simulation and SSI helps us understand the origin of biases and the total level of bias present in the data respectively. Improving on the existing image simulation tools and developing the pipelines to measure biases via SSI are necessary steps for future data releases with more stringent systematics requirements. My experiences with image simulation to validate blend removal and Abell 360 cluster measurements make cluster validation a natural next step that I am qualified to work on and lead.

With a validated pipeline in hand, we will have confident (or at least properly weighted) cluster weak lensing masses. For Rubin Data Preview 2 there are $\sim 100$ X-Ray detected clusters by eRosita and $\sim 400$ SZ detected clusters by ACT, which are more than enough to produce a LSST mass calibration for those external samples. Given that the previous ACT calibration uses a similar number of clusters [citation], it is possible to produce competitive results using this early data alone. Constraining cosmology from calibration constitutes another project that would require estimating a covariance, modeling shear profiles (either hierarchically or stacked), redshift distribution estimation, and running cosmology chains. As an active member in LSST DESC, I am excited to contribute to to those efforts with my skills and build towards a tomographic understanding of the cosmos.

*Goal 2: Mitigate weak lensing systematics*

I will continue to work on mitigating weak lensing systematics, especially blending. Extending my previous work on blend removal in [**unrec˙shear**] to include redshift is the next natural step towards understanding the effects of this methodology on cosmology. Completing a redshift study would enable a re-analysis of previous data, like DES Y3 or HSC Y3, providing a comprehensive look at the effect of unrecognized blends in weak lensing.

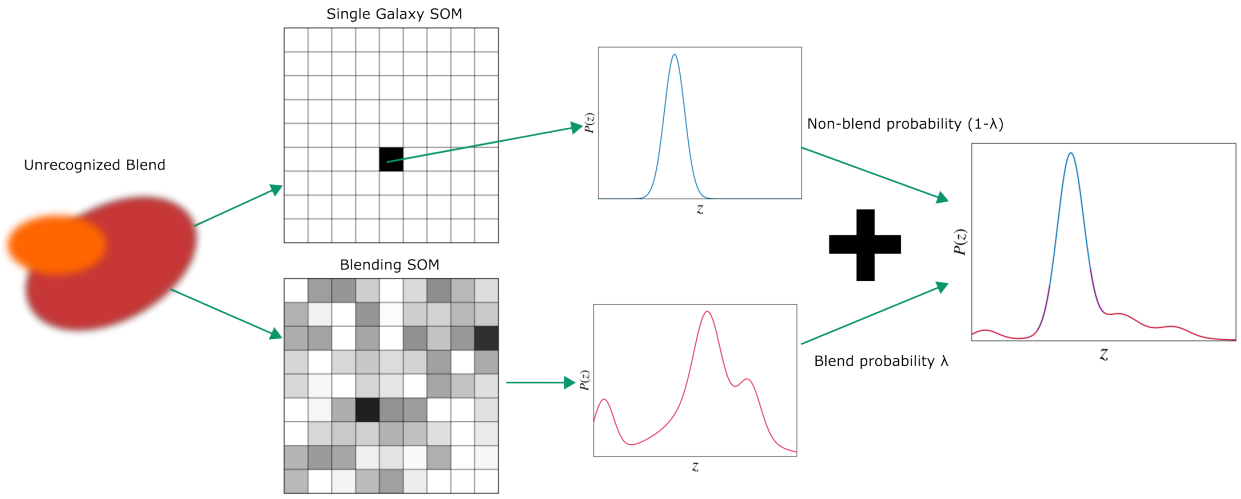



Figure 3: Example of modeling blends when producing redshift estimates via a SOM. Instead of only using the cell of the best fit, a weighted SOM can be made from synthetic blends that can be used to produce a more accurate PDF (color-coded to highlight contribution from individual terms).

While my previous work on blending has focused on removing these undetected objects, it may be possible to model the effects of blending for better mitigation without incurring a statistical cost. Using the machine learning probability from [**unrec˙liang**] as a prior, we can treat certain objects as a superposition of two galaxies which can lead to more accurate posteriors for redshift and shear.

The self-organizing map (SOM) is a powerful that projects the multidimensional color space down to a 2D surface and maintains the topology from the original data. The same map that is used to create ensemble redshift estimates can be extended to account for blends as outlined in Fig. 3. In that example, we use the blending probability to combine two redshift probability distributions: the distribution assuming a single galaxy, and the distribution assuming a superposition of two galaxies. In preliminary work, I have shown that this schema can produce more accurate posterior estimates than a single-galaxy SOM approach. For shears, accounting for cross-redshift blends can be done at the image processing level by improving deblending which can drive down shape noise. Both of these are relatively straight-forward to show as proof of concept and are promising ideas that can improve the quality of data used in cosmological analysis.

The best solution to account for unrecognized blends would be to recognize them which can be done through joint processing of space and ground based imaging. While there are strong solutions on how to best combine photometry, there is work to be done on combining shape measurements in order to maximize weak lensing signal. During my postdoc, I plan to contribute to the on-going efforts to enable joint-processing for both Metadetect and AnaCal. This will mostly rely on simulated datasets, like the existing Roman-Rubin simulations, but can be tested on data using deep field overlap of clusters fields which I am well equipped to study. If successful, this will completely circumvent the need for the machine learning techniques I have previously worked on wherever there is overlap between ground and space instrument. Those techniques may still find use detecting and mitigating unrecognized blends in space-based data of which there is no agreed upon solution.

This is an extremely exciting time to be doing cosmology and I have been at the forefront working with some of the first images taken by Rubin. I want to build on my experiences from my

PhD by working on weak lensing systematics and clusters so that we can better probe the universe tomographically.